

Research Statement Outline

GPAI-Driven Robotic World Models: Bridging System 1/2 for Smart Manufacturing

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1. Introduction and Vision

The future of Smart Manufacturing lies in robots that do not just follow static scripts but "reason" about their environment. My research vision focuses on the implementation of **General Purpose AI (GPAI)** that bridges the gap between intuitive perception (**System 1**) and analytical planning (**System 2**). By leveraging **State Space Models (SSMs)** and **Physics-Informed ML**, I aim to develop robotic "World Models" that can autonomously navigate the uncertainty of machine loading and human-robot collaboration in dynamic factory environments.

2. Proposed Research Axes

2.1. 1. Hybrid Perception-Planning Architectures (System 1/2)

I propose to investigate the integration of fast, reactive models (e.g., TCNs or Vision Transformers) for real-time scene understanding with slow, deliberative models (e.g., **Mamba-SSM** or **xLSTM**) for sequential decision-making. This dual-system approach allows the robot to handle high-frequency sensor data while maintaining a long-term "memory" of the manufacturing process, enabling flexible navigation and manipulation strategies that adapt to changing product types.

2.2. 2. Physics-Informed World Models for Robust Manipulation

Data-driven robotics often struggle with physical consistency. I propose to develop world models that incorporate the **kinematics and physical constraints** of industrial robots directly into the neural architecture using **Physics-Informed Neural Networks (PINN)**. By penalizing violations of physical laws during the "What-If" simulation phase, we can ensure that the GPAI-guided trajectories are not only efficient but also physically feasible and safe, reducing the gap between simulation (Webots/ROS) and reality.

2.3. 3. Explainable and Trustworthy Human-Centric AI

For robots to work alongside humans, their decisions must be interpretable. I propose to extend my work on **Gradient Saliency** and **Uncertainty Quantification** to the multimodal perception systems of (humanoid) robots. By providing real-time "Confidence Envelopes" and visual heatmaps of the robot's focus, we can create a transparent feedback loop for human operators, ensuring that the GPAI-driven navigation is both trustworthy and ethically aligned with safety standards in a shared manufacturing space.

3. Alignment with Bosch and HORIZON

My profile as an AI Engineer with a successful track record in **Hardware-Software Co-Design** and **Industrial Digital Twins** (Sonatrach) aligns perfectly with the requirements of Robert Bosch Manufacturing Solutions. My experience in achieving **0.04ms inference** for edge-based sequential models provides the technical foundation needed to implement these GPAI tools in

the ****ARENA2036**** high-speed manufacturing environment. I am eager to contribute to the HORIZON project's mission of defining the future of European Smart Manufacturing.